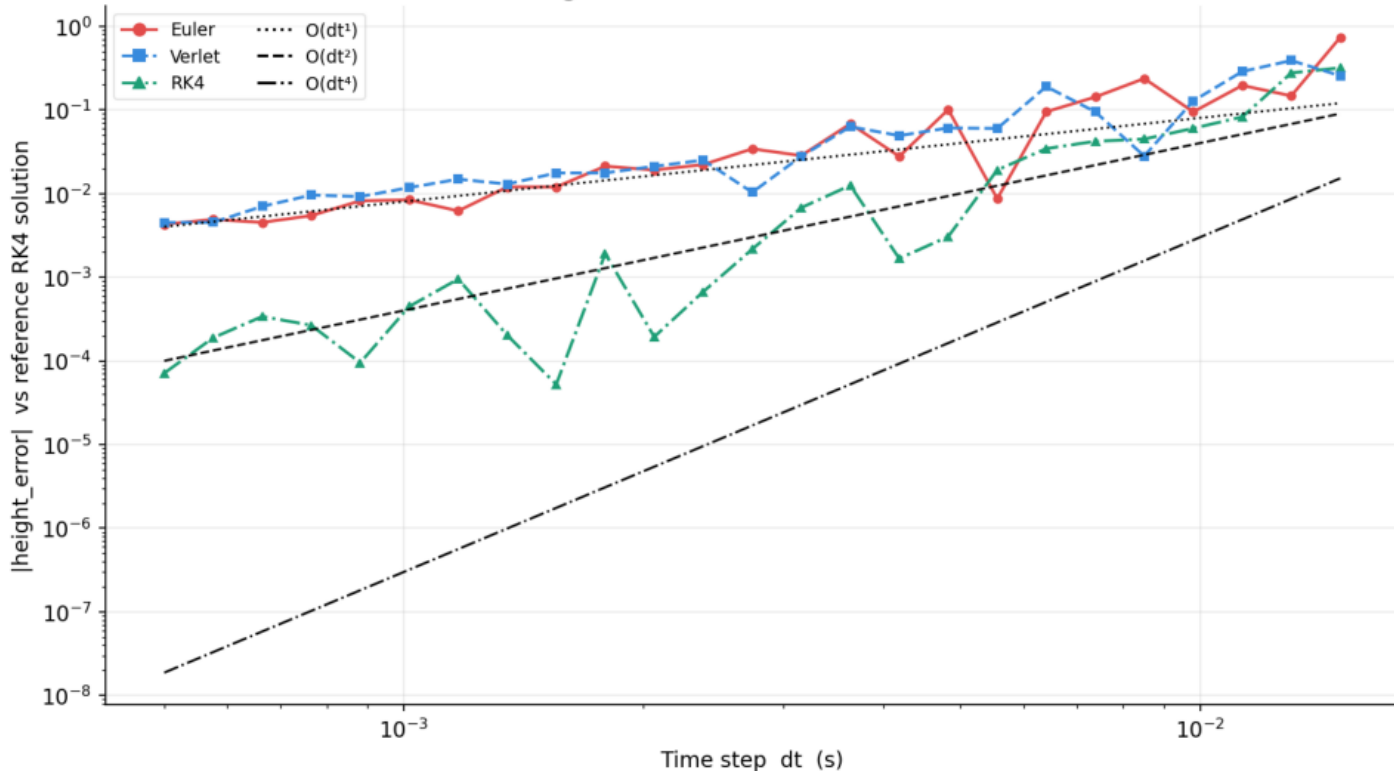


Peak height error vs dt — contact forces mode



CONVERGENCE — CONTACT FORCES

Unlike impulse-based contact, the spring-damper force is continuous → integrator order is partially visible.

Slopes (measured below):

Euler $\approx O(dt^{1.3})$

Verlet $\approx O(dt^{1.2})$

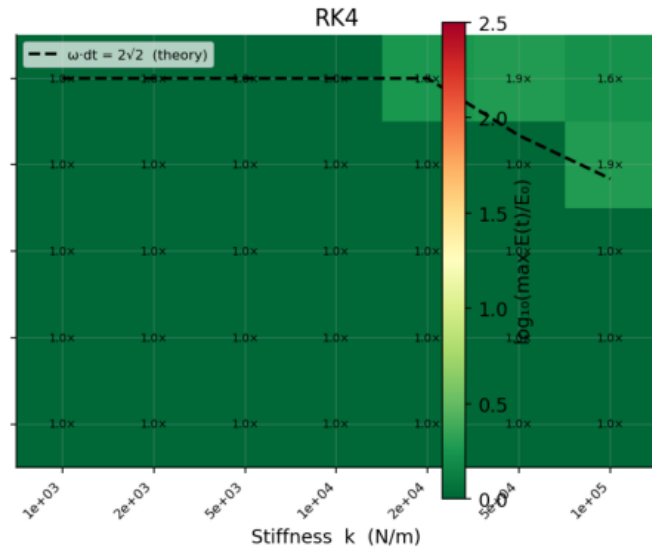
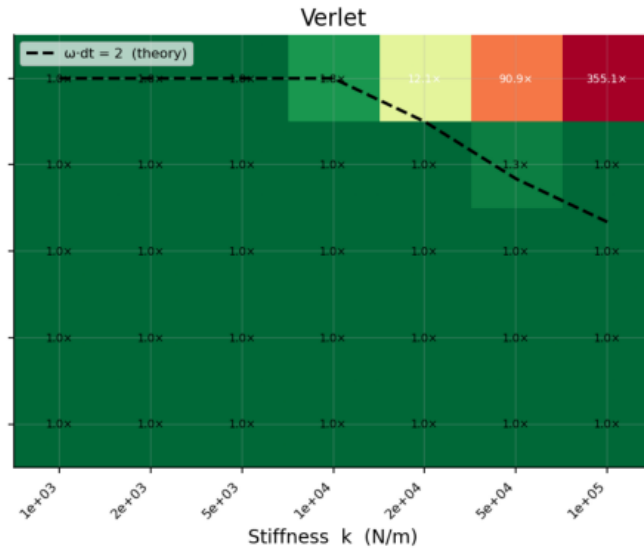
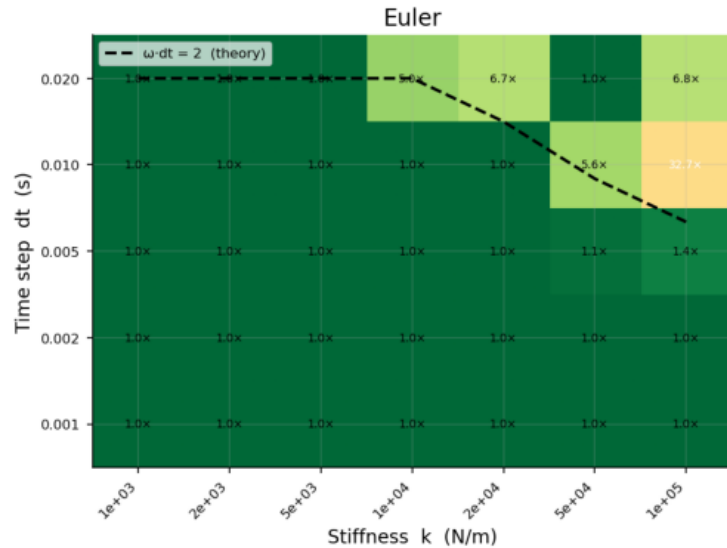
RK4 $\approx O(dt^{2.4})$

Why not $O(dt^2)$ / $O(dt^4)$?

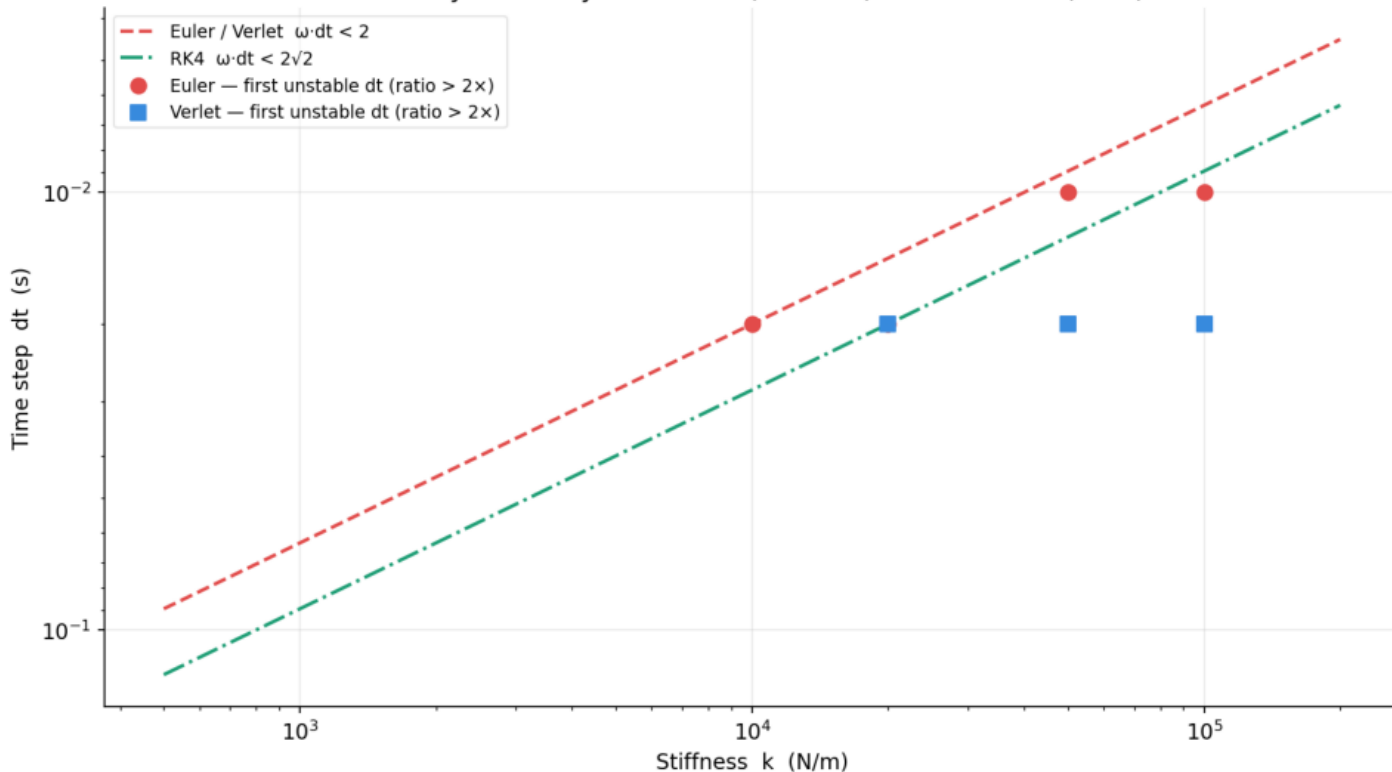
The $\max(F_{\text{spring}} + F_{\text{damp}}, 0)$ clamp introduces a C^0 discontinuity at contact onset/offset, capping the effective global order.

Despite this, RK4 is clearly the most accurate at any dt.

Stability map — max energy ratio $E(t)/E_0$ in the (k, dt) plane



Stability boundary: measured (markers) vs theoretical (lines)



STABILITY BOUNDARY

Theory (undamped spring):

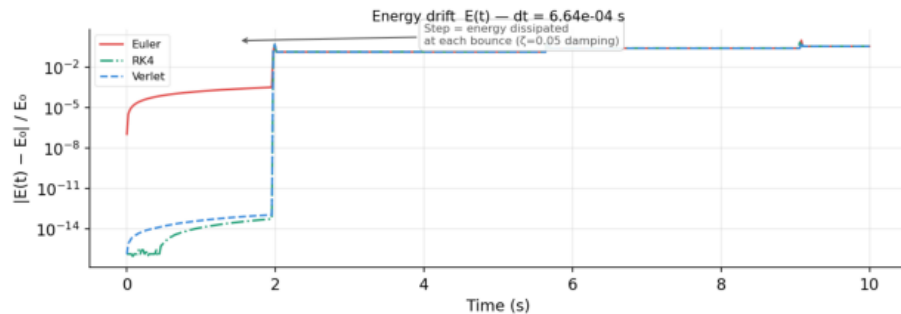
Euler/Verlet: $\omega \cdot dt < 2$

RK4: $\omega \cdot dt < 2\sqrt{2}$

RK4 can use a $\sqrt{2} \approx 41\%$ larger timestep than Euler/Verlet for the same stiffness k .

Practical implication:

For stiff contact (large k),
 RK4 allows coarser $dt \rightarrow$
 fewer steps $\rightarrow$ can be faster
 per unit of physical time,
 despite its $4\times$ cost per step.



ENERGY DRIFT $E(t) - \text{CONTACT FORCES}$

Staircase steps = energy lost at each bounce from viscous damping ($\zeta=0.05$). This is physics, not numerical error.

Between bounces (free flight):
Energy should be conserved.
Euler drifts slowly ($O(dt)$).
Verlet/RK4: near-exact.

The energy floor between steps is dt -dependent for Euler but solver-independent at the bounce steps (contact physics dominates there).

